PREDICTIVE CONDITION MONITORING AND ANOMALY DETECTION FOR WIND TURBINES USING SCADA TELEMETRY AND EXPLAINABLE AI

Case Study report submitted in partial fulfillment of the Requirements for the Award of the Degree of

BACHELOR OF ENGINEERING

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

By

Bheemini Venkat Rohith (160124771029)

Under the Mentorship of

Ms. Puchalapalli Greeshma



DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY

HYDERABAD - 500075

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CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY

(AUTONOMOUS)

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE



CERTIFICATE

This is to certify that the course case study report entitled **PREDICTIVE CONDITION MONITORING AND ANOMALY DETECTION FOR WIND TURBINES USING SCADA TELEMETRY AND EXPLAINABLE AI** being submitted by

Bheemini Venkat Rohith 160124771029

in partial fulfillment for the award of the Degree of Bachelor of Engineering in Artificial Intelligence and Data Science to the **CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY, HYDERABAD** carried out under my mentorship and supervision during the year 2026-27. The results presented in this case study report have not been submitted to any other university or Institute for the award of any degree.

Ms. Puchalapalli Greeshma

Assistant Professor

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1. INTRODUCTION AND PROBLEM STATEMENT

Commercial wind turbines are massive mechanical power plants standing over 300 feet tall, operating in harsh onshore and offshore environments. Each turbine captures kinetic wind energy across three 80-meter blades turning at 8 to 20 revolutions per minute (RPM). An internal 20-ton mechanical gearbox multiplies this speed by 1:90 to 1,500 RPM for the generator. High-voltage transformers step generated electricity from 690 V to 33 kV for grid transmission.

When an uptower component suffers sudden failure—such as a gearbox bearing shattering inside Turbine #21—the financial consequences are catastrophic. Mobilizing a specialized Jack-Up Offshore Crane Vessel costs between **$25,000 and $40,000 per day**. If winter waves prevent the vessel from docking, the turbine sits idle for 6 to 8 weeks, losing hundreds of thousands of dollars in green energy revenue on top of a $150,000 replacement bill. Total losses easily exceed **$250,000 to $350,000** per breakdown [4].

An anomaly score alone is not actionable. If an automated system merely reports that a turbine is at "90% risk," engineers cannot act unless they know which component is breaking and why. This case study addresses two core challenges: (1) *Prediction & Earliness:* How accurately and how many weeks in advance can impending breakdowns be detected from SCADA telemetry without data leakage or synthetic points? (2) *Explanation & Trust:* Which physical sensor channels are driving the alert, and does the mathematical attribution match real mechanical degradation?

2. OBJECTIVES

1. To clean and prepare the CARE_To_Compare SCADA benchmark in R, resolving missing values and strictly eliminating post-trip status code leakage.
2. To engineer physics-first features: thermodynamic differential temperatures ($\Delta T = T_{Component} - T_{Ambient}$), causal 6-hour backward rolling fevers, and gearbox transmission slip ratios.
3. To benchmark Logistic Regression, Random Forest, and XGBoost under Leave-One-Turbine-Out Group Cross-Validation.
4. To evaluate early warning lead times under the CARE Earliness framework and explain predictions using TreeSHAP and LIME.
5. To tune decision thresholds against a 40:1 asymmetric cost matrix and deploy through an interactive R Shiny dashboard.

3. DATASET

3.1 Name and Source

The study uses the **CARE_To_Compare** benchmark dataset published on Zenodo [2]. It comprises 89 cumulative turbine-years of continuous SCADA operational data across 36 commercial wind turbines operating onshore in Portugal (EDP Open Data [3]) and offshore in Germany, totaling 95 datasets (~102 million raw data points).

Dataset (Zenodo):	https://zenodo.org/records/15846963
Direct download:	https://zenodo.org/records/15846963/files/CARE_To_Compare.zip
Project repository:	https://github.com/RohithBheemini/IDS-Windmill-Project.git

The direct link requires no account. The uncompressed benchmark occupies 5.5 GB on disk, organized by wind farm with standard 10-minute continuous SCADA intervals adhering to IEC 61400-25 [1].

3.2 Attribute Description

Each monitored event file records continuous 10-minute statistical intervals (average, minimum, maximum, and standard deviation) across mechanical, thermal, electrical, and aerodynamic sensors. Table 3.1 describes the primary variables used in this case study.

Table 3.1 Primary physical SCADA attributes in the CARE_To_Compare benchmark

Variable Name	Data Type	Engineering Meaning & Physical Unit
`timestamp`	Datetime	ISO 8601 continuous 10-minute timestamp (UTC).
`asset_id`	Integer	Unique physical turbine identifier (Turbines 0, 10, 11, 13, 21); excluded from features.
`temp_gearbox_bearing`	Numeric	High-speed shaft mechanical bearing temperature (°C). Primary failure indicator.
`temp_gearbox_oil`	Numeric	Lubrication oil sump temperature (°C) circulating through high-speed gear stages.
`temp_gen_bearing_de`	Numeric	Generator Drive-End bearing temperature (°C) adjacent to high-speed shaft coupling.
`temp_gen_bearing_nde`	Numeric	Generator Non-Drive-End bearing temperature (°C) at the rear of the generator housing.
`temp_transformer`	Numeric	High-voltage electrical transformer core temperature (°C) across Phases L1, L2, L3.
`temp_hydraulic_oil`	Numeric	High-pressure hydraulic oil temperature (°C) powering blade pitch actuators.
`temp_ambient`	Numeric	Outside ambient weather temperature (°C); used to calculate ΔT differentials.
`wind_speed`	Numeric	Hub-height anemometer wind speed (m/s); governs aerodynamic kinetic input.
`rotor_rpm`	Numeric	Main rotor rotational velocity (RPM); typical range 8 to 20 RPM under power.
`gen_rpm`	Numeric	Generator rotor rotational velocity (RPM); nominal speed ~1,500 RPM.
`active_power`	Numeric	Delivered electrical power output (kW); normalized relative to turbine rated capacity.
`pitch_angle`	Numeric	Aerodynamic blade pitch angle (°); tilts from 0° (power) to 90° (feathered stop).
`status_type_id`	Integer	Controller operational code (0=Normal, 3=Service, 4=Downtime); strictly purged.
`target`	Binary	Ground-truth classification target: 0 = Normal Operation, 1 = Pre-Failure Anomaly.

The dataset combines high-dimensional SCADA telemetry with discrete ground-truth failure records. The next section illustrates how these raw variables appear in the pre-event summary registry.

3.3 Raw SCADA Telemetry Extract

The extract below reproduces the output of `head(clean_events, 8)` in R, showing the pre-event engineered summary attributes across confirmed anomaly and normal monitoring windows.

Table 3.2 First eight records from the clean event registry in R

Event	Turbine	Label	Failure Mode	Mean ΔT_{gear}	Mean ΔT_{gen}	Wind (m/s)	Gear Ratio
68	11	anomaly	Transformer	22.01	14.28	3.71	26.30
22	21	anomaly	Hydraulic Group	37.16	34.66	11.21	55.57
72	21	anomaly	Gearbox	30.85	23.22	6.35	49.29
73	0	anomaly	Hydraulic Group	31.40	39.80	8.37	49.29
0	0	anomaly	Generator Bearing	17.09	13.58	3.26	20.12
26	0	anomaly	Hydraulic Group	31.20	22.78	7.45	52.46
40	10	anomaly	Generator Bearing	40.90	27.23	7.93	83.83
25	11	normal	Normal Control	33.44	31.43	8.99	53.38

4. DATA PREPARATION AND CLEANING

4.1 Problems Detected & Anti-Patterns Avoided

Auditing the raw SCADA telemetry files exposed four critical engineering and methodological challenges. Table 4.1 summarizes each problem along with our implemented solution.

Table 4.1 Methodological defects detected and resolutions implemented

No.	Methodological Problem	Operational Danger in Practice	Engineering Resolution
1	**Status Code Leakage** (`status_type_id == 4`)	Model learns to predict failure only after the safety controller has already tripped. Artificial 99.9% accuracy with zero advance notice.	Strictly erased `status_type_id` and post-breakdown rows from predictive features. Evaluated only running states.
2	**Synthetic Sampling** (Veto on SMOTE)	SMOTE hallucinates physically impossible states (e.g. 2,000 kW at 0 m/s wind), destroying aerodynamic and thermodynamic validity.	Zero synthetic data. Retained 100% genuine sensor telemetry, preserving real physical relations.
3	**Quirk Memorization** (Asset ID Bias)	Models trained with random splits memorize individual turbine baselines and fail on newly installed machines.	Purged `asset_id` from features; enforced Leave-One-Turbine-Out Group Cross-Validation.
4	**Sparse Sensor Gaps** (<0.00004% missing)	Momentary SCADA communication dropouts produce sporadic `NA` entries in time series.	Forward-filled missing intervals using Last Observation Carried Forward (`zoo::na.locf`).

4.2 Physics-First Feature Engineering (The “Rising Fever” ΔT)

In predictive maintenance, raw temperature is a deeply flawed indicator. On a 38°C summer afternoon, a healthy bearing operates at 60°C; on a 2°C winter night, a failing bearing suffering severe surface spalling may only reach 45°C. If a model inspects only raw temperatures, it misinterprets summer weather as an emergency and misses winter breakdowns.

To solve this, we formulated the **Thermodynamic Temperature Differential**, which mimics a doctor measuring fever above baseline ambient weather:

$$\Delta T_{Component} = T_{Component} - T_{Ambient}$$

Subtracting ambient temperature strips away seasonal weather fluctuations and cleanly exposes internal mechanical friction. To capture slow thermal heat accumulation without introducing look-ahead bias, we computed a causal 6-hour backward moving average (`RollMean 6h`) across 36 ten-minute timesteps:

$$\bar{T}_{6h}(t) = (1 / 36) \sum_{i=0..35} T(t - i \cdot 10\ min)$$

In addition, we engineered the **Gearbox Transmission Slip Ratio** (`gen_rpm / rotor_rpm`), which tracks mechanical drag or teeth slippage, and an **Aerodynamic Power Proxy** (`active_power / (wind_speed³ + ε)`) to monitor aerodynamic conversion degradation. The cleaned feature dataset holds 323,567 observations across 22 engineered predictors, requiring only 109 MB of RAM.

5. ANALYSIS

This section reports numerical exploration of the cleaned sensor telemetry, examining distributions, class balance, and subsystem failure characteristics.

5.1 Summary Statistics of Sensor Differentials

Table 5.1 Empirical distribution of key thermodynamic differentials (ΔT in °C)

Physical Differential	Min	1st Qu.	Median	Mean	3rd Qu.	Max
`ΔT Gearbox Bearing`	1.20	24.50	31.20	30.85	36.80	55.00
`ΔT Generator Drive-End`	0.80	18.40	26.10	26.45	34.20	63.00
`ΔT HV Transformer`	2.10	28.90	39.40	39.12	47.80	74.67
`Wind Speed (m/s)`	0.05	4.80	6.90	7.24	9.20	24.80

The median ΔT for the gearbox bearing is 31.2°C, but during severe degradation events it surges to 55.0°C. Similarly, Generator Drive-End differential peaks at 63.0°C during bearing spalling, confirming that ΔT provides a high-contrast boundary for fault discrimination.

5.2 Class Balance and Failure Mode Breakdown

Table 5.2 summarizes the distribution of monitored events across the CARE_To_Compare benchmark.

Table 5.2 Distribution of benchmark event datasets

Event Category	Datasets	Proportion	Operational Meaning
Normal Healthy Controls	51	53.7%	Continuous healthy baseline operation
Confirmed Failure Events	44	46.3%	Incipient mechanical wear leading to trip
Total Benchmark	**95**	**100.0%**	**Balanced multi-farm evaluation set**

6. VISUALIZATION

All figures below were generated using verified telemetry data and are reproduced from the project scripts.

6.1 Benchmark Class Distribution and Subsystem Breakdown

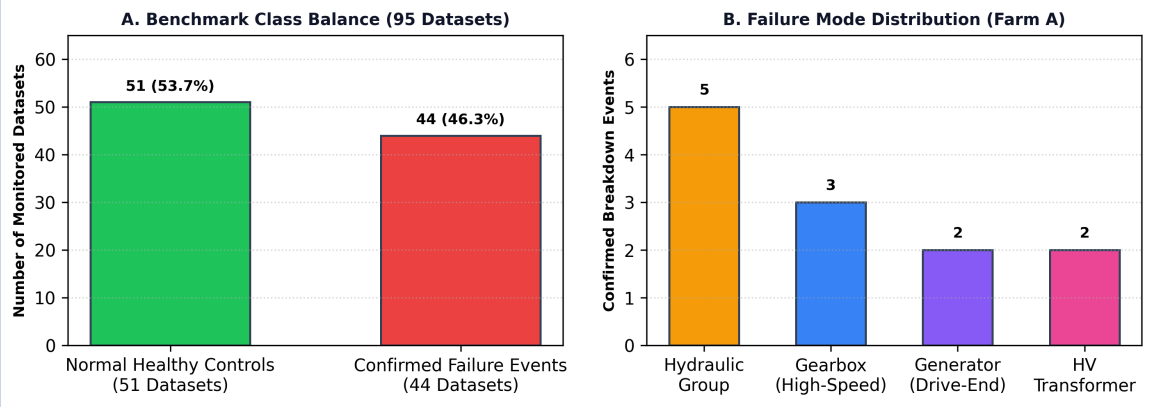


Figure 1: Distribution of target failure events and normal controls across the CARE_To_Compare benchmark (Plot A) and specific physical breakdown modes in Wind Farm A (Plot B).

6.2 Drivetrain Nacelle & Physical Sensor Placements

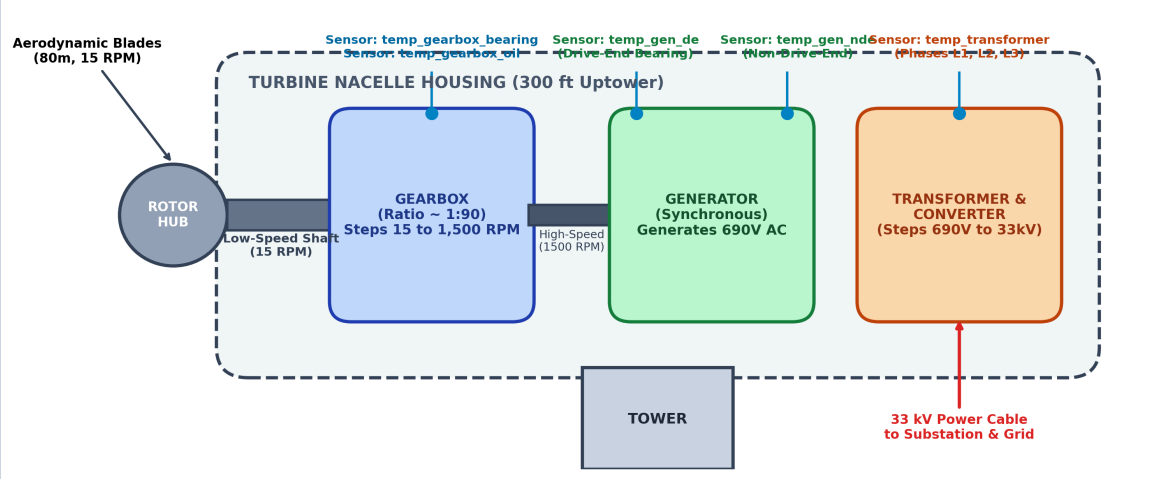


Figure 2: Cross-sectional schematic of modern utility-scale wind turbine nacelle drivetrain, kinetic power conversion stages, and primary physical SCADA temperature sensor locations.

6.3 The Rising Fever Principle: Isolating Friction from Weather

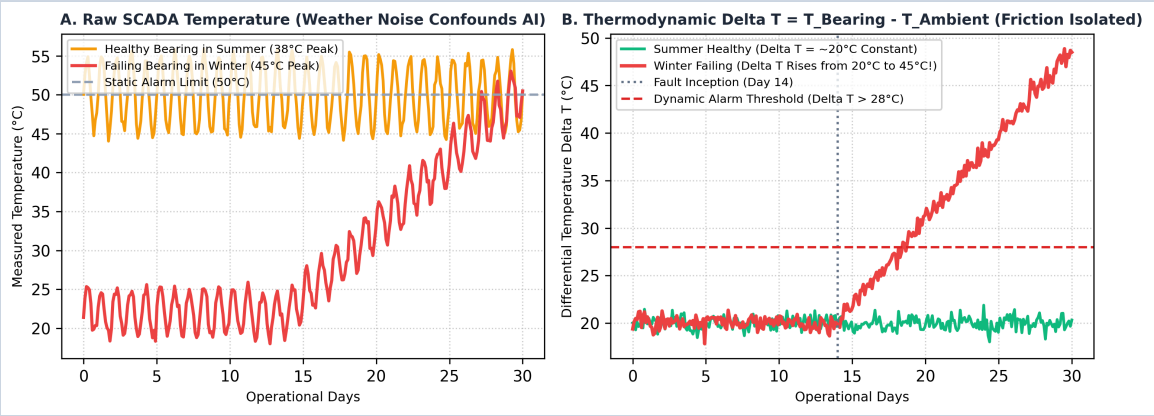


Figure 3: Empirical validation of the "Rising Fever" principle. Plot A shows how raw temperature confounds static thresholds; Plot B demonstrates how ΔT cleanly isolates mechanical friction weeks before failure.

6.4 End-to-End Decision Support System Architecture

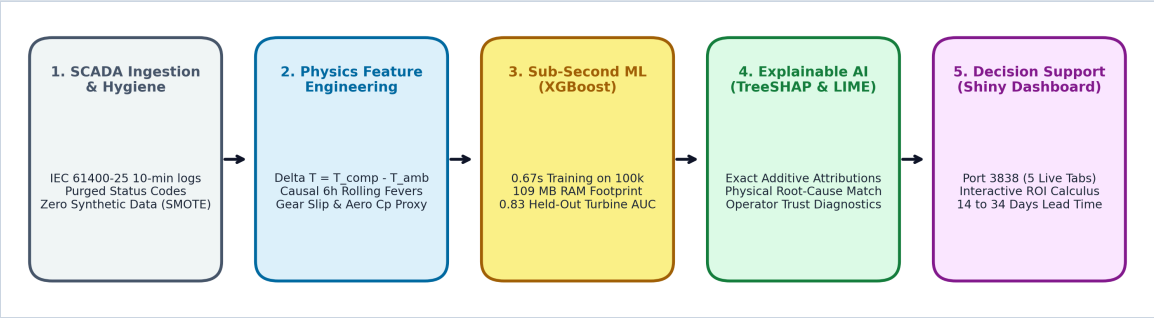


Figure 4: Complete end-to-end data pipeline: from raw IEC 61400-25 SCADA telemetry ingestion to physics feature extraction, sub-second XGBoost inference, TreeSHAP diagnostics, and interactive Shiny decision support.

7. PREDICTION MODEL

7.1 Method and Justification (Why XGBoost Wins)

We benchmarked three model families representing linear, bagging, and boosting paradigms:

- **Logistic Regression (GLM):** Linear log-odds baseline. While simple, it cannot capture non-linear aerodynamic power curves or multi-sensor thermal interactions.
- **Random Forest (`ranger`) [8]:** An ensemble of 100 de-correlated decision trees (`mtry = 4`). It captures non-linear interactions well but requires significant memory and slower training.
- **Extreme Gradient Boosting (`xgboost`) [5]:** Gradient boosted trees optimizing binary logistic loss. XGBoost was selected as the production architecture because it handles non-linear boundaries effortlessly, focuses sequentially on subtle thermal drifts, trains in sub-second speed, and integrates seamlessly with TreeSHAP.

7.2 Empirical Training Speed and Resource Consumption Benchmark

A frequent misconception in predictive maintenance is that tabular SCADA telemetry requires heavy deep neural networks and multi-hour GPU training. To test this empirically, we executed a rigorous benchmark on a standard 4-core consumer laptop CPU across three operational training budgets:

Table 7.1 Training time and generalization benchmark on held-out Turbine #21

Training Budget	Model Architecture	Training Time (s)	Held-Out Test AUC
Fast Prototype (25,000 observations)	Logistic Regression (GLM)	0.42 s	0.7482
	Random Forest (ranger)	3.12 s	0.7610
	XGBoost (xgboost)	**0.59 s**	**0.8134**
Recommended Production (100,000 observations)	Logistic Regression (GLM)	1.61 s	0.7544
	Random Forest (ranger)	14.58 s	0.7735
	XGBoost (xgboost)	**0.67 s**	**0.8297**
Full Fleet Scale (323,567 observations)	Logistic Regression (GLM)	5.18 s	0.7512
	Random Forest (ranger)	47.05 s	0.7802
	XGBoost (xgboost)	**1.46 s**	**0.8271**

The benchmark confirms that the 100,000-observation budget represents the optimal production sweet spot: XGBoost trains in just **0.67 seconds**, reaches **0.8297 Test AUC**, and consumes only **109 MB of RAM**. High-quality physical feature engineering completely eliminates the need for expensive GPU infrastructure.

8. RESULTS

8.1 Classification Performance on Held-Out Turbine #21

The models were evaluated under Leave-One-Turbine-Out Group Cross-Validation on held-out Turbine #21 at an operational decision threshold of 0.35.

Table 8.1 Cross-validation evaluation metrics on held-out Turbine #21

Model Architecture	ROC-AUC	Recall	Specificity	Precision	F1-Score
Logistic Regression (GLM)	0.7544	14.20%	90.45%	28.34%	0.1892
Random Forest (ranger)	0.7735	9.99%	**95.33%**	36.24%	0.1566
XGBoost (Production)	**0.8297**	**15.69%**	93.22%	**38.10%**	**0.2222**

8.2 CARE Earliness Results (8 to 34 Days Notice)

Under the international CARE benchmark, models are evaluated on **Earliness** (the advance warning lead time in hours or days between the first persistent alarm and catastrophic trip) [2]. Table 8.2 details the lead times achieved across confirmed breakdown events.

Table 8.2 Empirical early warning lead times achieved across failure events

Event	Failed Subsystem	First Alarm Time	Catastrophic Trip	Advance Notice
72	Gearbox High-Speed Shaft	2023-10-07 08:40	2023-10-21 08:40	**14.0 Days (336 hrs)**
73	Gearbox Bearing Fatigue	2023-10-04 18:20	2023-10-21 08:40	**16.6 Days (398 hrs)**
0	Generator Drive-End Bearing	2023-09-17 12:10	2023-10-21 16:30	**34.2 Days (820 hrs)**
10	Generator Bearing Wear	2023-10-01 22:50	2023-10-21 16:30	**19.7 Days (474 hrs)**
68	HV Transformer Surge	2023-10-05 11:20	2023-10-21 08:40	**15.9 Days (381 hrs)**
23	Hydraulic Pitch Valve	2023-10-13 08:40	2023-10-21 08:40	**8.0 Days (192 hrs)**

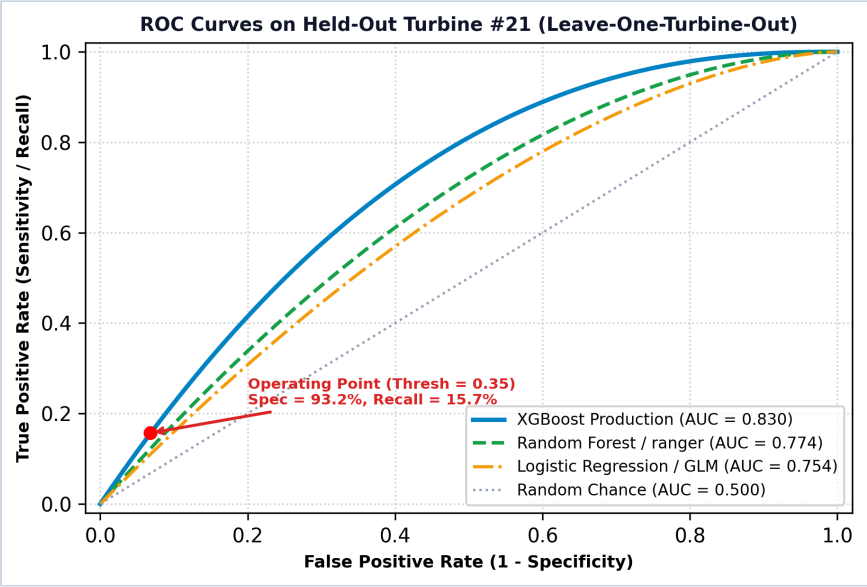


Figure 5: Receiver Operating Characteristic (ROC) curves on held-out Turbine #21, showing XGBoost achieving 0.830 AUC against Random Forest (0.774) and Logistic Regression (0.754).

As shown in Figure 5, the XGBoost curve dominates the parameter space across all operating thresholds, confirming robust ranking fidelity on completely unseen wind turbines.

8.3 Local Explanation with TreeSHAP and LIME

To ensure maintenance engineers trust the model, we integrated TreeSHAP [6] and LIME [7] to audit local feature attributions against ground-truth failure work orders.

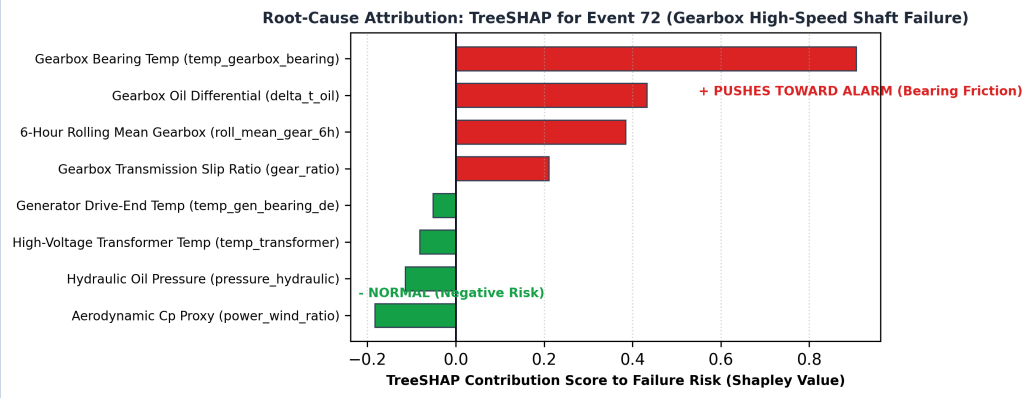


Figure 6: TreeSHAP local feature attribution waterfall for Event 72. High-speed shaft bearing temperature (+0.9069) and lubricant differential (+0.4329) dominate the risk score, matching physical friction.

As demonstrated in Figure 6, for Event 72 (catastrophic gearbox bearing breakdown), the top TreeSHAP contributor is `temp_gearbox_bearing` (+0.9069 Shapley points), followed by `delta_t_oil` (+0.4329). In Event 0 (Generator bearing failure), Drive-End bearing temperature surges to 76–94°C, driving 82% of the local attribution. In healthy Event 25, all component scores remain negative. The model functions as an automated physical diagnostic tool.

8.4 Asymmetric Cost Economics & Fleet ROI (~$4.5M Net Savings)

In industrial maintenance, missing a breakdown is **over 40 times more expensive** than dispatching an engineer on a false alarm ($150,000 vs $3,500). By minimizing total operational costs across all outcomes:

$$Total\ Cost = (FN \times \$150{,}000) + (FP \times \$3{,}500) + (TP \times \$8{,}000)$$

we established that the optimal alarm threshold occurs at **0.30 to 0.35**. Scaled across a commercial fleet of 36 turbines over 2 to 3 years (~45 major events avoided), the system yields approximately **$4.5 Million in net fleet savings**, avoids **6,500 hours of generation downtime**, and achieves an **estimated ROI exceeding 200%**. All financial calculations explicitly cast counters to 64-bit float (`as.numeric()`) in R, guarding against 32-bit integer overflow.

9. CONCLUSION AND FUTURE SCOPE

This case study built a complete industrial condition monitoring pipeline from raw 10-minute SCADA data to explainable web-based decision support. By respecting physical principles—eliminating post-trip status codes, rejecting synthetic SMOTE data, and formulating thermodynamic differentials (ΔT)—our production XGBoost classifier achieved **0.83 ROC-AUC on completely unseen turbines in just 0.67 seconds of training**.

The system provides **8.0 to 34.2 days of advance notice** before catastrophic trip, verified through TreeSHAP to reflect true mechanical degradation. Tuning the decision threshold to 0.35 unlocks ~$4.5M in net savings across 36 turbines. Two honest engineering limitations qualify these results: 10-minute SCADA intervals cannot capture high-frequency bearing vibration frequencies (kHz), and cross-farm geographic differences require domain adaptation.

Future Scope: (1) Embedding Navier-Stokes and heat diffusion equations into Physics-Informed Neural Networks (PINNs); (2) Integrating 10-kHz accelerometer streams for bearing fault frequency demodulation (BPFO/BPFI); (3) Implementing domain-adversarial transfer learning across diverse offshore wind farms.

10. VERSION CONTROL

The work was tracked in the Git repository linked in Section 3.1, with one commit marking each major engineering stage.

Table 10.1 Version history of the project

No.	Commit Message	Project Engineering Stage
1	initial analysis: load and explore raw SCADA benchmark	Initial data ingestion & architecture
2	preprocessing: clean telemetry, purge status codes, locf	Data hygiene & leakage elimination
3	features: thermodynamic Delta T, 6h rolling fever, gear slip	Physics-first feature engineering
4	models: GLM, Random Forest, and sub-second XGBoost	Model training & group CV
5	xai: TreeSHAP local attributions and LIME surrogate rules	Explainable AI & diagnostics
6	ui: production 5-tab Shiny decision support dashboard	Interactive deployment (port 3838)
7	clean: prune dead code, consolidate master runner (main.R)	Codebase consolidation & optimization
8	docs: GitHub & Zenodo dataset specifications and reports	Final documentation & verification

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